VEHICLE SERVICE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

SINDU MG

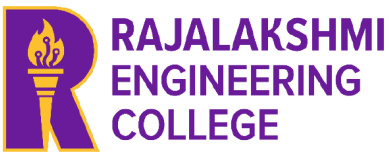
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in partial fulfilment of the End Semester Examination for the course

CB23F35 – LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY
COMPUTER SCIENCE AND BUSINESS SYSTEMS



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(An Autonomous Institution, Affiliated to Anna University, Chennai)
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BONAFIDE CERTIFICATE

Certified that this project report titled “VEHICLE SERVICE MANAGEMENT SYSTEM” is the bonafide work of SINDU MG (241401101), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, SINDU MG (241401101), hereby declare that the project report titled “VEHICLE SERVICE MANAGEMENT SYSTEM”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its form, reports, dashboard, approval process and workflow — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 03 November 2026

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SINDU MG

ABSTRACT

Many small and medium vehicle service centres still record service bookings on paper job cards, in spreadsheets or through phone calls and messaging groups. Customer details, vehicle details and the progress of each job are kept in separate places, so the service advisor spends time re-entering data, customers are not informed reliably when a booking is accepted, and the manager has no live view of how many vehicles are booked, in progress or completed. Because the entries are free text, errors such as a wrongly typed registration number or a vehicle model entered in the wrong field go unnoticed.

This project presents the Vehicle Service Management System, a low-code application built on Zoho Creator. A single form, Vehicle_Service_Form, organised into Customer Details, Vehicle Details and Service Details sections, captures the customer name, contact number, e-mail address, vehicle number, brand, model and fuel type of every booking. Two list reports present the stored records, and a Service Dashboard shows Total Customer, Service Booking, In progress and Completed status panels, a pie chart of service status and an embedded report. A form workflow, Notify customer, sends a confirmation e-mail with the subject “Record Added” to the customer on every successful submission, and an approval process, Customer Approval, routes each new booking to a Level 1 approver and e-mails the customer when it is approved.

The report documents the application and validates it against the supplied screenshots. Five bookings are listed in the reports, one approval request was completed on 01 November 2026 and no requests are pending. The validation review shows that typed fields and the mandatory customer name are enforced, but it also finds a registration number (TN214567) that does not follow the Indian registration-mark format, a body type (“SEDANS”) stored as a vehicle model, a fuel type that should be confirmed with the customer, and dashboard panels that count four customers while the reports hold five records. Each finding is matched with a correction that can be made in the builder, including a proposed Deluge validation script. The report keeps configured features separate from features whose operation was observed.

Keywords: Low-Code Development; Zoho Creator; Vehicle Service Management; Approval Workflow; Data Validation; Deluge Scripting; Dashboard Reporting

PROJECT AT A GLANCE

Attribute	Details
Project Title	VEHICLE SERVICE MANAGEMENT SYSTEM
Application Domain	Automobile after-sales – service booking, customer and vehicle records, booking approval and status tracking
Student Name & Roll No.	SINDU MG – 241401101
Platform	Zoho Creator (trial edition; the application header shows “Trial expires in 3 days” on 01 November 2026)
Application Name	Vehicle Service management System (short name shown in the menu: VSM)
Account / Approver ID	sindu.mg.2024.csbs_rajala599 (Level 1 approver, Figure 5.8); live-mode user name SINDU
Demo Credentials	Provided to the examiner at the demonstration (not printed in this report)
Number of Forms / Subforms	1 (Vehicle_Service_Form, with three sections) / none
Number of Reports (by type)	2 – List: Vehicle_Service_Form Report, Customer report
Pages / Dashboards	1 – Service Dashboard (four KPI panels, a pie chart and an embedded report)
Workflows	Form: 1 (Notify customer – Send mail on successful form submission) | Approval: 1 (Customer Approval) | Schedule: none
Deluge Functions and Scripts	None in the application; actions are configured visually. A proposed validation script and function are given in Appendix C
Integrations	None configured
Roles & Permission Profiles	Not configured in the supplied material (platform defaults)
Records at Testing	5 service bookings; approvals: 0 pending, 1 completed

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
BR	Business Rule
CMVR	Central Motor Vehicles Rules
CNG	Compressed Natural Gas
CRUD	Create, Read, Update, Delete
DMS	Dealer Management System
EV	Electric Vehicle
FR	Functional Requirement
KPI	Key Performance Indicator
NFR	Non-Functional Requirement
RTO	Regional Transport Office
UI	User Interface
URL	Uniform Resource Locator
VSM	Vehicle Service Management (System)

CHAPTER 1
INTRODUCTION

1.1 Background

A vehicle service centre receives cars and two-wheelers for periodic maintenance, repairs and inspections. For every vehicle the centre needs the customer’s name and contact details, the registration number, brand, model and fuel type of the vehicle, the service requested and the progress of the job. The service advisor, the workshop and the manager each need a different view of the same information, and the customer expects to be told when the booking is accepted and when the vehicle is ready.

In a manual process these steps are handled through paper job cards, spreadsheets, phone calls and messages. This project builds the Vehicle Service Management System on Zoho Creator, a low-code platform, so that booking entry, approval, customer notification, status tracking and reporting are handled in one connected application, and so that the data entered can be checked at the point of entry.

1.2 Problem Statement

There is no single place in which a customer’s details, the vehicle details, the booking decision and the service status are connected. As a result:

- Customer and vehicle details are written on job cards and copied into spreadsheets, which leads to duplication and inconsistent records.
- There is no structured approval trail showing which bookings have been accepted and when.
- Customers are not informed automatically when a booking is recorded or approved.
- The manager cannot see the number of customers, bookings in progress or completed services without preparing a summary by hand.
- Free-text entries are not validated, so a wrong registration number, an incomplete phone number or a model typed in the wrong field remains in the records.

1.3 Objectives

- To design a form that captures customer, vehicle and service details with appropriate field types.
- To provide an approval process through which every new booking is reviewed by a designated approver.
- To notify the customer by e-mail when a booking is submitted and when it is approved, using workflows.

- To provide reports and a dashboard that give a live view of customers, bookings and service status.
- To document the application and validate its data and behaviour against the evidence, and to propose corrections for the gaps that are found.

1.4 Scope of the Project

In scope: one form (Vehicle_Service_Form) with Customer Details, Vehicle Details and Service Details sections; two list reports; one dashboard page; one approval process; one form workflow that sends an e-mail; and a validation review of the stored data. The application is used through the Zoho Creator web interface in live mode.

Out of scope for the present work, because it is not shown in the project material: spare-parts inventory, invoicing and payment, technician allocation, scheduled reminders for the next service, integration with other Zoho services and a customer portal. Some of these are listed as future enhancements in Section 9.2.

1.5 Justification for a Low-Code Approach

Low-code development suits this problem for the following reasons [1], [2]. First, the data of a service booking is form-centred, which maps directly to the visual form builder of Zoho Creator. Second, the approval and the customer e-mail are configured visually in the Workflow section, without writing server code, which shortens development time. Third, hosting, sign-in and the mail service are provided by the platform, so no separate server has to be maintained. Finally, because the services offered and the vehicle brands handled by a centre change often, an application that can be modified quickly by a non-specialist developer is appropriate.

1.6 Technology Stack Summary

The application is assembled entirely from platform components. Table 1.1 lists each layer of the solution and the way it is used in this project, so that the reader can relate every later chapter to a concrete technology.

Table 1.1: Technology Stack of the Application

Layer	Technology	Use in this Project
Application platform	Zoho Creator (trial edition)	Hosts the form, reports, dashboard page, workflow and approval
Data capture	Form with sections	Vehicle_Service_Form: Customer Details, Vehicle Details, Service Details
Presentation	List reports; dashboard page	Vehicle_Service_Form Report, Customer report; Service Dashboard with KPI panels, pie chart and embedded report

Layer	Technology	Use in this Project
Process control	Approval process; form workflow	Customer Approval (Level 1 approver); Notify customer on successful form submission
Notification	Zoho Creator mail service (Send mail action)	Confirmation e-mail “Record Added” and approval e-mail to the customer’s address
Scripting	Deluge	Not used in the present version; proposed validation script and function in Appendix C
Access	Zoho account sign-in	Application opened by the signed-in owner (SINDU)
Client	Web browser on a desktop computer	Used to build, test and capture the screenshots

1.7 Organisation of the Report

Chapter 2 reviews the existing system and related literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator, with screenshots placed beside the feature they show. Chapter 6 covers testing, the data validation review and observations. Chapter 7 discusses deployment, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a conventional service centre the customer calls or walks in, and the service advisor writes the customer's name, phone number and vehicle details on a job card. The job card travels with the vehicle through the workshop, and the status of the job is known only to the people who handle it. At the end of the day the advisor may copy the bookings into a spreadsheet, and the customer is informed by phone when the vehicle is ready.

2.2 Limitations of the Existing System

- Customer and vehicle data is written more than once, so the same vehicle may have conflicting entries.
- There is no approval trail or status tracking for bookings.
- Customers are informed only if someone remembers to call or message them.
- Summaries such as the number of vehicles in progress must be prepared by hand and are not available in real time.
- There is no validation of entries such as the registration number, phone number or e-mail address.

2.3 Technology Background

Zoho Creator. A low-code platform for building web and mobile business applications. Applications are built from forms, reports, pages, workflows and settings, and are hosted by the platform [2].

Approvals and workflows. Zoho Creator lets the developer attach an approval process to a form, name the approvers for each level and define the actions to run on Approve and Reject; form workflows run actions such as Send mail when a record is created or edited [2], [4].

Deluge. The scripting language of Zoho Creator. It is used in workflows and functions for validating input, fetching records, applying conditions and sending mail [3].

Registration marks. In India the registration mark of a vehicle consists of the state code, the code of the registering authority, an optional series of letters and a number of up to four digits, as prescribed under the Central Motor Vehicles Rules, 1989 [7]. This format is used in Chapter 6 to validate the stored vehicle numbers.

2.4 Literature Survey

Table 2.1: Summary of Literature Survey

No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	W. Rosam and O. Kalnichevski, BPB Publications [1]	Low-Code Development: A Practical Guide	Explains low-code concepts, benefits and limits, and how business users and developers share the work	Basis for the justification in Section 1.5 and the limitations in Section 8.4
2	Zoho Corporation, online documentation [2]	Zoho Creator Help Documentation	Describes forms, reports, pages, workflows, approvals and settings of Zoho Creator	Reference for every component built in Chapter 5
3	Zoho Corporation, online documentation [3]	Deluge Script Reference Guide	Describes the syntax and tasks of Deluge, including validation and sendmail	Reference for the proposed scripts in Appendix C
4	Zoho Corporation, learning resources [4]	Zoho Creator Learning Centre	Guided tutorials and examples of applications built in Zoho Creator	Guided the form, report, dashboard and workflow design
5	Y. Luo, P. Liang, C. Wang, M. Shahin and J. Zhan, 2021 [5]	Characteristics and Challenges of Low-Code Development: The Practitioners' Perspective	Analyses practitioner discussions and reports the characteristics and challenges of low-code development	Informs the limitations in Section 8.4
6	M. A. Al Alamin et al., 2021 [6]	An Empirical Study of Developer Discussions on Low-Code Software Development Challenges	Analyses developer discussions about low-code and the challenges they raise	Informs the limitations in Section 8.4
7	Ministry of Road Transport and Highways, Government of India [7]	Central Motor Vehicles Rules, 1989 – Rule 50	Prescribes the form and manner of display of registration marks on motor vehicles	Basis for the vehicle number validation rule in Section 6.4

2.5 Proposed System

The proposed system is a Zoho Creator application named Vehicle Service management System. It keeps customer, vehicle and service details in one form with sections, so each booking is entered once (limitation 1). An approval process gives the approver a Pending Tasks and Completed Tasks list with the date and time of each review (limitation 2). A form workflow sends a confirmation e-mail to the customer when a booking is submitted, and the approval sends a second e-mail when the booking is approved (limitation 3). The Service Dashboard and the reports give live counts instead of hand-made summaries (limitation 4). Typed fields and a mandatory customer name provide a first level of validation, and Chapter 6 proposes further rules (limitation 5).

2.6 Comparison of Service Management Approaches

Three broad approaches are available to a service centre: continuing with job cards and spreadsheets, adopting a ready-made dealer management system (DMS), or building a small custom application on a low-code platform. Table 2.2 compares them on the aspects that mattered for this project. The comparison is qualitative and is based on the general characteristics described in the documentation and literature cited above [2], [5]. For a small centre the low-code application offers most of the structure of a DMS at a much lower cost and can be changed by the centre itself, which is why it was chosen.

Table 2.2: Comparison of Three Approaches to Vehicle Service Management

Aspect	Job cards and spreadsheets	Ready-made DMS	Low-code application (this project)
Set-up effort	None, but every step is manual	Licence, set-up and training on the product	Form, reports and workflow built once in the builder
Fields and forms	Free-form columns; no enforced types	Fixed by the product, with limited custom fields	Designed to the centre’s needs, with typed fields
Approval and status tracking	None; depends on the advisor	Provided by the product	Approval process with Pending and Completed Tasks; status chart
Customer notification	Phone calls and messages	Product-defined	Send mail on submission and on approval
Reporting	Prepared by hand	Product reports	List reports and a dashboard from stored records
Data validation	None	Product-defined	Field types and mandatory rules; Deluge validation can be added
Cost for a small centre	Low	Usually high	Low; platform subscription after the trial

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Stakeholder	Role in the Application
Customer (vehicle owner)	Provides customer and vehicle details for the booking; receives the confirmation e-mail and the approval e-mail at the address entered in the Email field.
Service Advisor / Administrator	Enters bookings through Vehicle_Service_Form, reviews them under Approvals and updates the service status.
Level 1 Approver	The account sindu.mg.2024.csbs_rajala599, named as the Level 1 approver of Customer Approval (Figure 5.8). Approves or rejects each new booking.
Service Manager	Monitors the Service Dashboard and the reports. No separate role is configured; in testing the owner account performs this task.
Application owner (SINDU)	Builds the application and acts as advisor, approver and manager during testing; shown as the signed-in user in every live-mode screenshot.

3.2 Functional Requirements

Table 3.2: Functional Requirements

ID	Requirement
FR1	The system shall capture the customer name (first and last name), contact number with country code and e-mail address.
FR2	The system shall capture the vehicle number, vehicle brand (from a list), vehicle model and fuel type (Petrol, Diesel or Electric).
FR3	The system shall capture service details, including a service status of Booked, In Progress or Completed.
FR4	The system shall send a confirmation e-mail to the customer when a booking is submitted successfully.
FR5	The system shall create an approval request for each new booking, list it under Pending Tasks and, after review, under Completed Tasks, and e-mail the customer when the booking is approved.
FR6	The system shall list all bookings and customers in reports with search and add-record options.
FR7	The system shall provide a dashboard showing total customers, bookings, jobs in progress, completed jobs and a chart by service status.

ID	Requirement
FR8	The system shall reject incomplete or wrongly formatted entries before a record is stored.

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

ID	Requirement
NFR1	Usability: the form shall be grouped into sections and use suitable field types (name, phone, e-mail, drop-down, radio).
NFR2	Data quality: dashboard totals shall be computed from the stored records and agree with the reports.
NFR3	Availability: the application shall be hosted by Zoho Creator and reachable through a web browser.
NFR4	Maintainability: fields, reports and workflows shall be changeable in the builder without redeploying code.
NFR5	Security: only signed-in users of the account shall open the application, and only the named approver shall approve bookings.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Item	Requirement
Hardware	A computer with an internet connection
Browser	A modern web browser
Platform	Zoho Creator (trial edition)
Mail	A customer mailbox to receive the confirmation and approval e-mails

3.5 Use Case Model

Table 3.5: Use Case Description – Book a Vehicle Service

Item	Description
Actor	Service Advisor (on behalf of the customer)
Pre-condition	The advisor is signed in and opens Vehicle_Service_Form → Vehicle_Service_Form.
Main flow	1. The advisor enters the customer, vehicle and service details. 2. The advisor clicks Submit. 3. The record is stored, the Notify customer workflow sends “Record Added” to the customer and an approval request is created.

Item	Description
Alternative flow	If the customer name is empty or the phone or e-mail is wrongly formatted, the form shows an error and the record is not stored.
Post-condition	The booking is listed in the reports and counted on the dashboard.

Table 3.6: Use Case Description – Approve a Booking

Item	Description
Actor	Level 1 Approver
Pre-condition	A booking is waiting under Approvals → Pending Tasks.
Main flow	1. The approver opens Pending Tasks. 2. The approver checks the booking details. 3. The approver clicks Approve. 4. A mail is sent to the customer's e-mail and the request moves to Completed Tasks.
Alternative flow	The approver clicks Reject; no action is configured for Reject in the present version.
Post-condition	The reviewed request is listed under Completed Tasks with the date and time of review.

3.6 Business Rules and Assumptions

The rules in Table 3.7 were identified from the form, workflow and approval settings of the application. Chapter 6 states which of them were actually observed.

Table 3.7: Business Rules and Assumptions

ID	Rule or Assumption	Source in the Application
BR1	The customer name must be entered before a booking can be stored.	Mandatory option in the builder, Figure 5.3
BR2	The contact number is entered with a country code (+91 by default).	Phone field, Figure 5.2
BR3	The vehicle brand is selected from a list, and the fuel type is one of Petrol, Diesel or Electric.	Drop-down and radio fields, Figure 5.2
BR4	Every successful submission sends "Record Added – Service booking submitted successfully!" to the customer's e-mail.	Workflow Notify customer, Figures 5.11–5.13
BR5	Every new booking is approved by the Level 1 approver, and the customer is e-mailed on approval.	Approval Customer Approval, Figure 5.8
BR6	The service status of a booking is one of Booked, In Progress or Completed.	Pie chart legend, Figure 5.7
A1	The owner account acts as advisor, approver and manager during development and testing.	Signed-in user SINDU in every screenshot
A2	The Zoho Creator trial remains active while the application is demonstrated.	Trial banner shows 3 days on 01 November 2026

3.7 Feasibility Study

Technical feasibility. Every required capability — typed forms with sections, an approval process, an e-mail workflow, reports and a dashboard — is provided by Zoho Creator. The application was built and used through an ordinary browser, which shows that no special hardware or server installation is needed.

Operational feasibility. The form follows the fields already written on a job card, and the approval list works like an inbox, so advisors need little training. Customers do not need to sign in; they only receive e-mails.

Economic feasibility. The application was developed on a trial account, so no licence cost was incurred during the course. Continued use after the trial would require a paid plan, which is recorded as a risk in Section 7.4.

Schedule feasibility. The completed approval task carries the date 01 November 2026, and the application was in use with five records on that date, which shows that a working first version can be completed within the time available for the project.

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The application runs entirely on Zoho Creator. Users work in live mode through the menu; the form stores bookings; the form workflow and the approval process react to form events; the platform mail service sends the e-mails; and the reports, the dashboard and the approval lists read the stored records. The structure is shown in Figure 4.1.

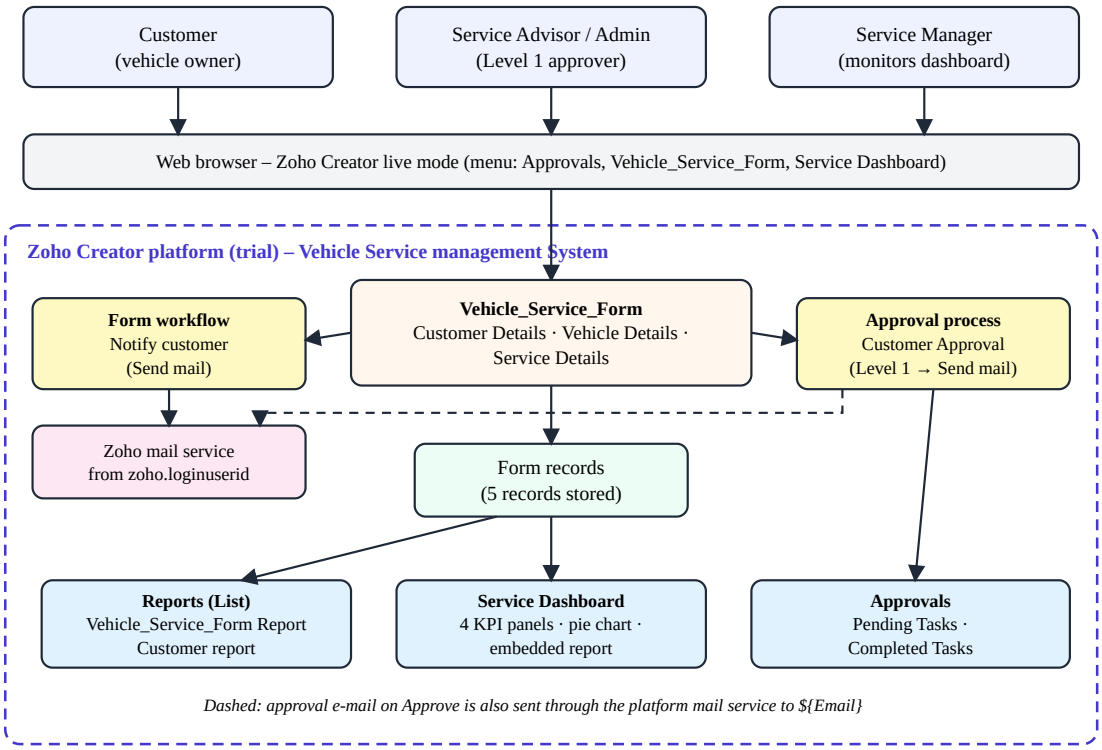


Figure 4.1: System Architecture of the Application

The approval e-mail is drawn with a dashed arrow because the supplied material shows the action configured but does not show a received approval e-mail.

4.2 Data Model

The application has one form, Vehicle_Service_Form, which is the only data table. Each record holds one booking and contains the customer, the vehicle and the service in three sections. Because a returning customer with a second vehicle would be stored again with the same name, phone and e-mail, the design is simple but not normalised; a separate Customer form linked by a lookup field is proposed in Section 9.2.

Table 4.1: Form and Its Purpose

Form / Section	Purpose	Related Reports and Pages
Vehicle_Service_Form – Customer Details	Captures who the booking belongs to and how to contact them	Vehicle_Service_Form Report, Customer report
Vehicle_Service_Form – Vehicle Details	Identifies the vehicle to be serviced	Both reports; Service Dashboard embedded report
Vehicle_Service_Form – Service Details	Records the service requested and its status	Service Dashboard KPI panels and pie chart; Approvals

4.3 Form Design

The field specification below is taken from the live form (Figure 5.2), the form builder (Figure 5.3) and the column values in the reports. The Service Details section is visible as a heading in the builder, but its fields are below the visible area of the screenshots; the status field is inferred from the dashboard legend.

Table 4.2: Field Specification – Vehicle_Service_Form

Section	Field Label	Field Type	Req.	Remarks
Customer Details	Customer_Name	Name	Yes	First Name and Last Name shown; Prefix and Suffix not displayed
Customer Details	Contact_Number	Phone	No	Country code +91 with flag; placeholder 81234 56789
Customer Details	Email	Email	No	Used as the recipient ${Email} of both e-mails
Vehicle Details	Vehicle_Number	Single Line	No	Free text; no format rule
Vehicle Details	Vehicle_Brand	Drop Down	No	Values seen: Maruti Suzuki, Hyundai, Tata
Vehicle Details	Vehicle_Model	Single Line	No	Free text, e.g. Swift, i20, Nexon
Vehicle Details	Fuel_Type	Radio	No	Petrol, Diesel, Electric
Service Details	Service status (inferred)	Drop Down / Radio	—	Values Booked, In Progress, Completed seen in the dashboard legend; exact field name not visible

4.4 Process and Workflow Design

The intended process is shown in Figure 4.2. A submitted booking is checked by the field rules, stored, confirmed to the customer by e-mail and sent to the approver; on

approval the customer receives a second e-mail; the service status then moves from Booked to In Progress to Completed, and the reports and dashboard reflect the change. The figure is a design view; the evidence for each step is presented in Chapter 5 and the observations in Chapter 6.

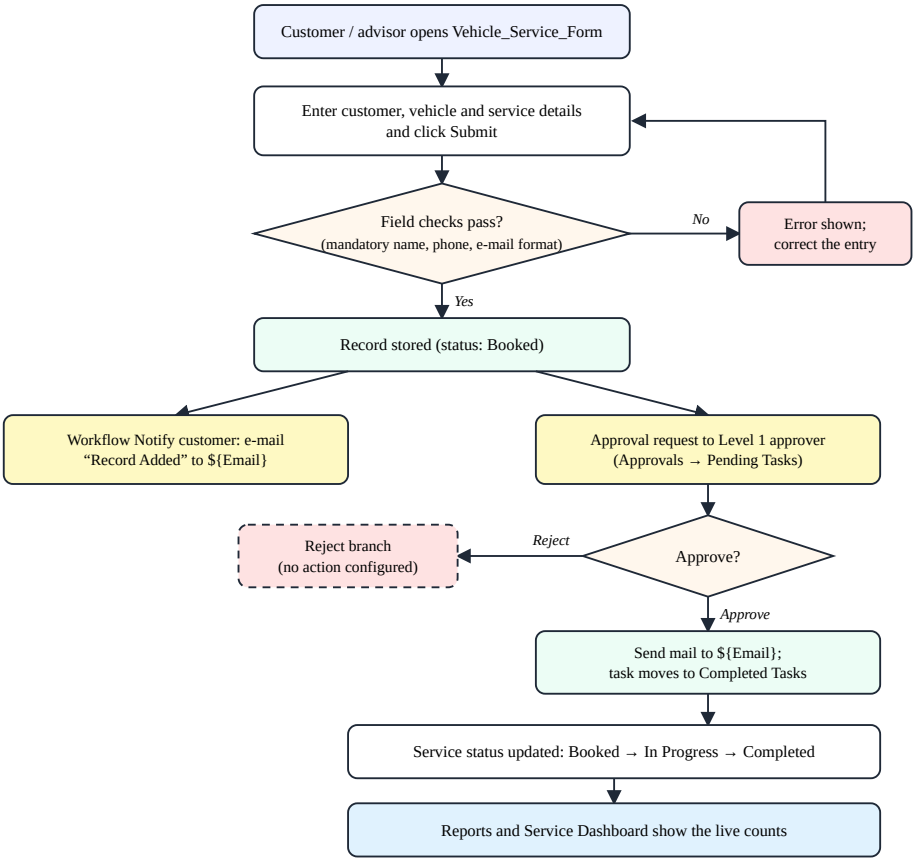


Figure 4.2: Workflow / Process Flow of a Service Booking

Table 4.3: Design of the Workflow and Approval

Automation	Trigger	Action	Evidence in this Report
Notify customer (form workflow)	Vehicle_Service_Form → Created → Successful form submission; Execute without condition	Send mail from zoho.loginuserid to ${Email}; subject "Record Added"; message "Service booking submitted successfully!"; Plain Text	Configuration shown in Figures 5.11–5.13; received e-mail not shown
Customer Approval (approval process)	Records added in Vehicle_Service_Form	Actions on start: none. Level 1 approver sindu.mg.2024.csbs_rajala599. On Approve: Send mail to ${Email}. On Reject: not visible	Configuration in Figure 5.8; one completed task in Figure 5.10

4.5 User Interface and Navigation Design

The left-hand menu of the application, with the VSM badge and the title Vehicle Service management System, is organised into three sections: Approvals (Pending Tasks and Completed Tasks), Vehicle_Service_Form (the form itself, Vehicle_Service_Form Report and Customer report) and Service Dashboard. The signed-in user name SINDU is shown at the bottom of the menu. A dark navy menu with an indigo highlight for the selected item is used as the theme. The header provides desktop, tablet and phone preview icons, Share and Edit this application.

4.6 Access Control Design

The present version relies on the platform's sign-in: the application is opened by the signed-in owner, and approval rights are restricted to the account named as Level 1 approver. No custom roles, permission profiles or data-sharing rules are shown in the supplied material. Section 9.2 proposes Advisor, Manager and Customer roles.

4.7 Data Integrity and Validation Considerations

Because the title of this work stresses validation, the design measures that keep the stored data usable are listed here and tested in Section 6.4.

- **Typed fields.** Name, Phone and Email field types are used, so the platform checks the structure of the name, the phone number and the e-mail address on submission.
- **Mandatory field.** Customer_Name is mandatory (Figure 5.3), which prevents a booking without a customer.
- **Controlled values.** Vehicle_Brand is a drop-down and Fuel_Type is a radio group, so their values are consistent across records.
- **Free-text fields and duplicates.** Vehicle_Number and Vehicle_Model are single-line text with no pattern or list, and no "No duplicate values" rule is shown on Vehicle_Number, so wrong or repeated values are not caught; this is where the errors in Section 6.4 occur.
- **Recipient dependency.** Both e-mails are sent to ${Email}, but Email is not mandatory, so a booking without an e-mail address cannot be notified.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application, Vehicle Service management System, was created in Zoho Creator. The menu described in Section 4.5 is visible on the left of every live-mode screenshot. Figure 5.1 shows the application in live mode with the Vehicle_Service_Form section expanded, listing the form and its two reports, together with the Approvals section and the Service Dashboard. The header shows the trial banner “Trial expires in 3 days”.

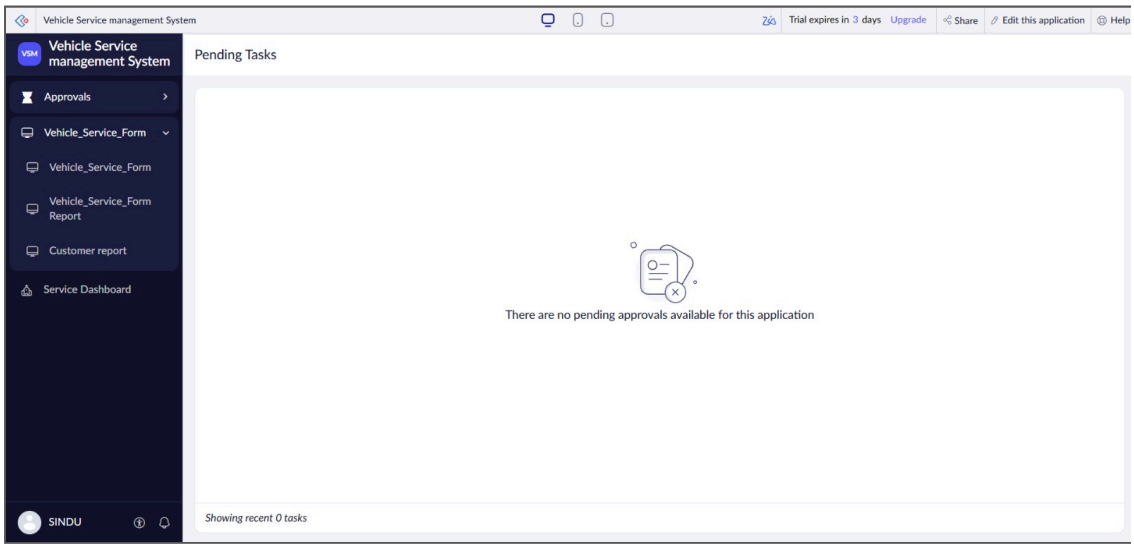


Figure 5.1: Application in Live Mode – Navigation Menu

5.2 Forms and Data Capture

5.2.1 Vehicle_Service_Form in Live Mode

Vehicle_Service_Form is the data-entry form of the application. Figure 5.2 shows it in live mode. Under **Customer Details** it collects Customer_Name (First Name and Last Name boxes, marked mandatory with an asterisk), Contact_Number with the Indian flag and the +91 country code, and Email. Under **Vehicle Details** it collects Vehicle_Number, Vehicle_Brand (a drop-down showing “-Select-”), Vehicle_Model, and Fuel_Type as radio buttons with the options Petrol, Diesel and Electric. The heading of the **Service Details** section begins at the bottom edge of the screenshot.

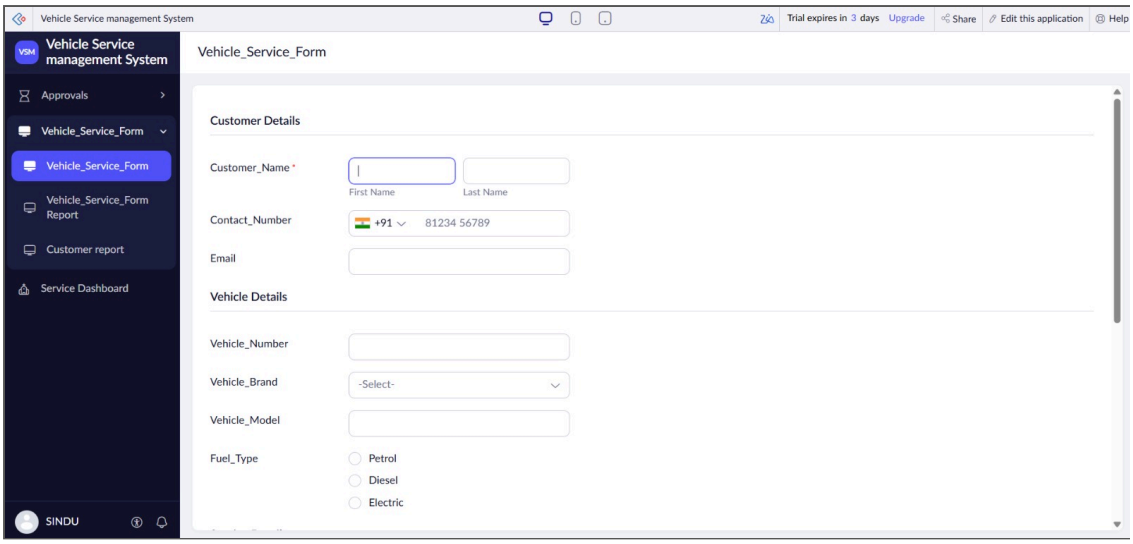


Figure 5.2: Vehicle_Service_Form in Live Mode (Customer Details and Vehicle Details)

5.2.2 Vehicle_Service_Form in the Form Builder

Figure 5.3 shows the same form in the form builder. The left panel lists the Basic Fields used to build it (Name, Email, Address, Phone, Single Line, Multi Line, Number, Date, Time and Drop Down). The canvas shows the three sections, Customer Details, Vehicle Details and Service Details, with the field icons indicating each type. The Field Properties panel on the right shows the selected field Customer_Name: its field name and field link name are both Customer_Name, the **Mandatory** validation option is ticked, and among the display fields First Name and Last Name are enabled while Prefix and Suffix are not.

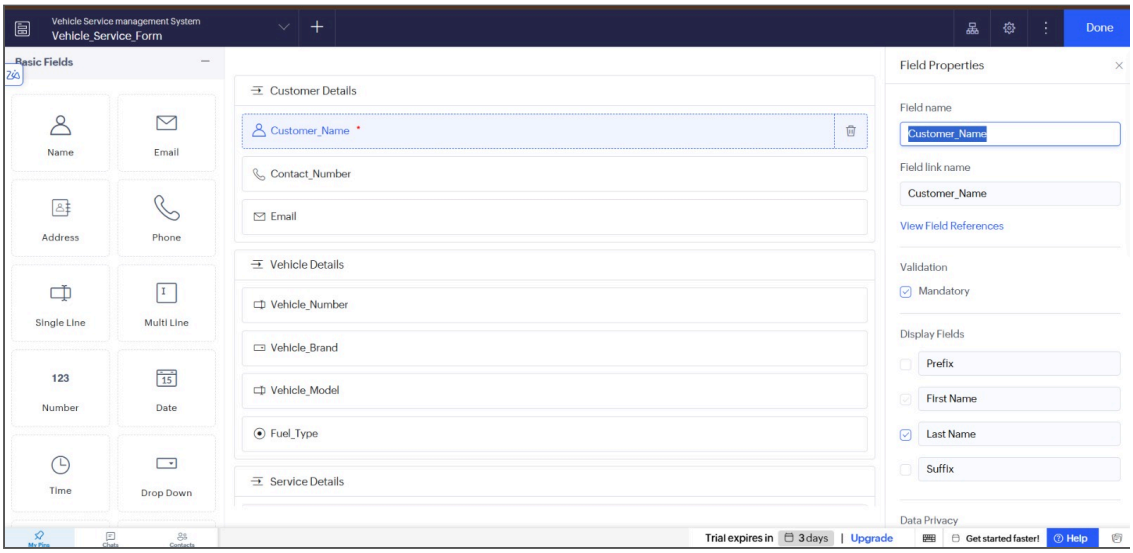


Figure 5.3: Vehicle_Service_Form in the Form Builder (Field Properties of Customer_Name)

5.3 Reports

Reports present the stored records. The two reports visible in the screenshots are summarised in Table 5.1.

Table 5.1: Reports Implemented

Report	Type	Source Form	What It Shows
Vehicle_Service_Form Report	List	Vehicle_Service_Form	Customer_Name, Contact_Number, Email, Vehicle_Number, Vehicle_Brand, Vehicle_Model, Fuel_Type (5 records)
Customer report	List	Vehicle_Service_Form	The same seven columns and the same five records

The Vehicle_Service_Form Report (Figure 5.4) lists every booking with the customer's contact details and vehicle details. Phone numbers are shown as clickable call links and e-mail addresses as mail links. The toolbar provides search, add record (+) and more options, and the footer reads "Showing 5 of 5".

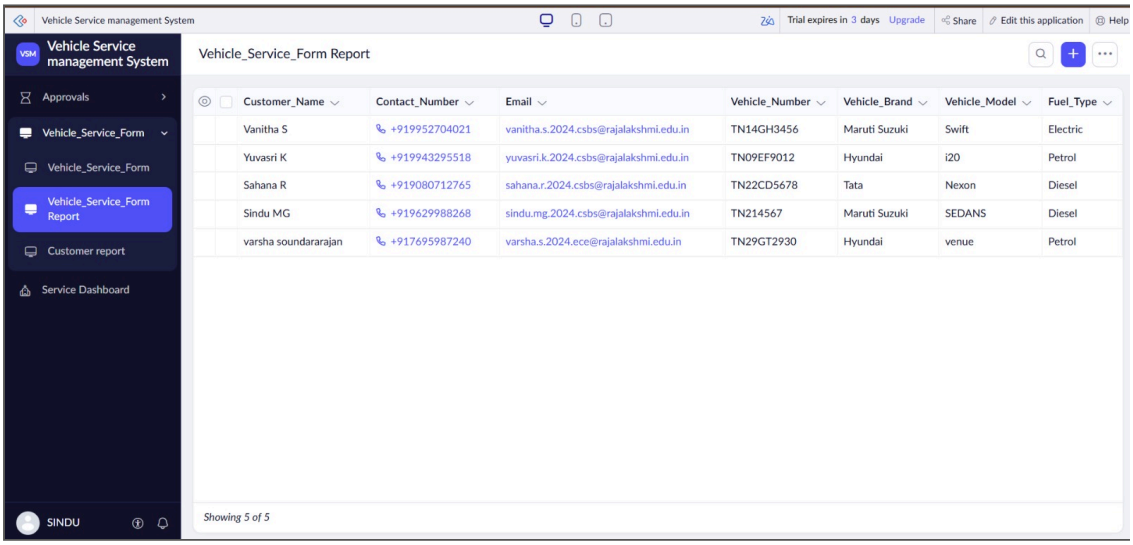


Figure 5.4: Vehicle_Service_Form Report (List)

The Customer report (Figure 5.5) is a second list report on the same form. It currently shows the same columns and the same five records as the Vehicle_Service_Form Report. Its name suggests a customer-focused view, so it is recommended in Table 6.4 that it show only the customer columns, or group the bookings by customer.

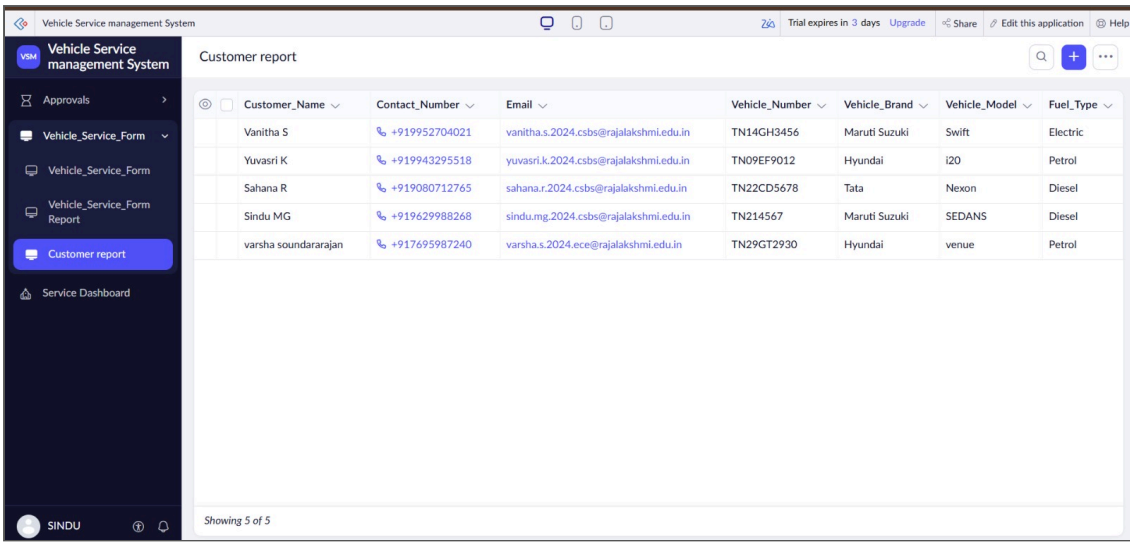


Figure 5.5: Customer Report (List)

5.4 Dashboard

The Service Dashboard is a page built with the page builder. Its upper part (Figure 5.6) shows four KPI panels in coloured cards: **Total Customer** (4), **Service Booking** (2), **In progress** (1) and **Completed status** (1). The panels for Total Customer and Service Booking carry icons.

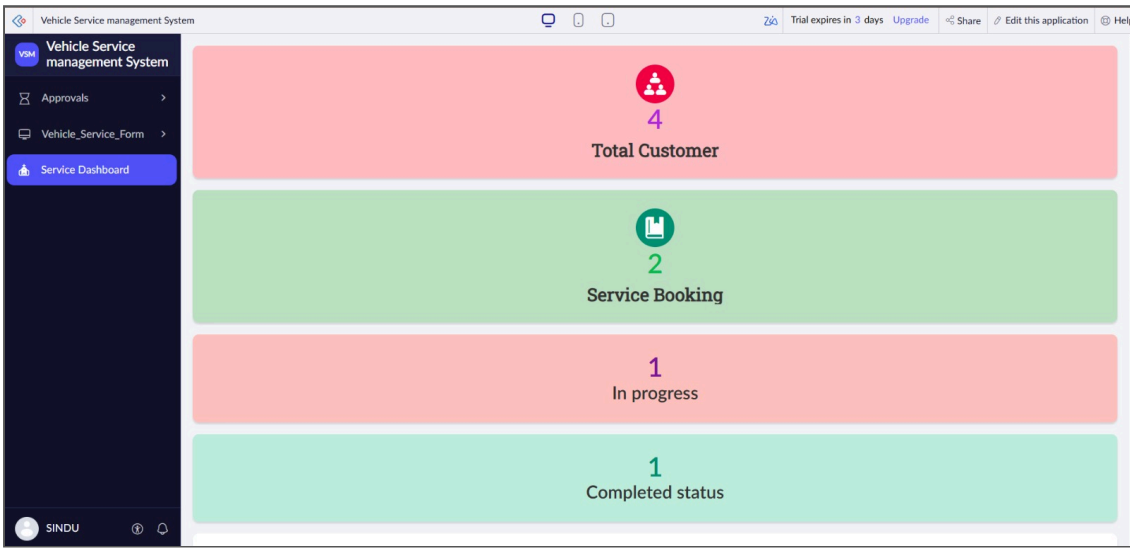


Figure 5.6: Service Dashboard – KPI Panels

Further down the page (Figure 5.7) a pie chart shows the bookings by service status with the legend Booked, In Progress and Completed. The Booked slice covers about three-fifths of the circle and the other two slices about one-fifth each, which corresponds to 3, 1 and 1 of the five stored records. Below the chart the page embeds the Vehicle_Service_Form Report, which shows the five bookings and the footer “Showing 5 of 5”. The difference between the KPI total (4) and the five records is examined in Section 6.5.

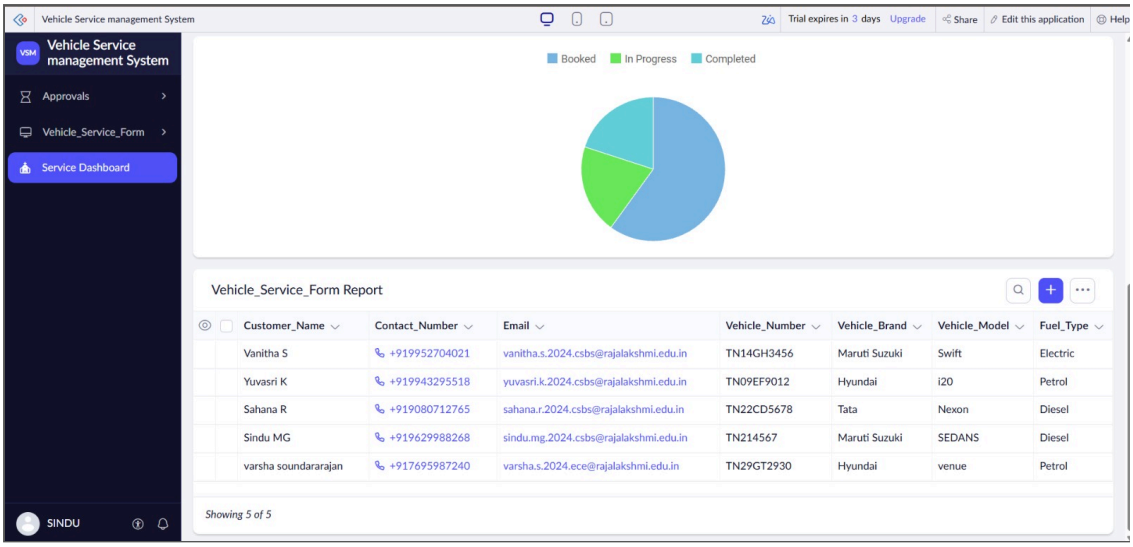


Figure 5.7: Service Dashboard – Service Status Pie Chart and Embedded Report

5.5 Approvals

An approval process named **Customer Approval** is attached to Vehicle_Service_Form (Figure 5.8). The page states “Below approvers at corresponding levels will approve the records added in the form Vehicle_Service_Form”. No action is defined under *Actions to execute on start*. The Level 1 approver is the account sindu.mg.2024.csbs_rajala599. Under **Approve** the action *Execute without condition → Send mail to ${Email}* is configured, so the customer is e-mailed when the booking is approved. A **Reject** branch is present at the bottom of the page; its actions are not visible in the screenshot. The Approval Events panel on the right lists On Start and, for the Level 1 approver, Approve and Reject.

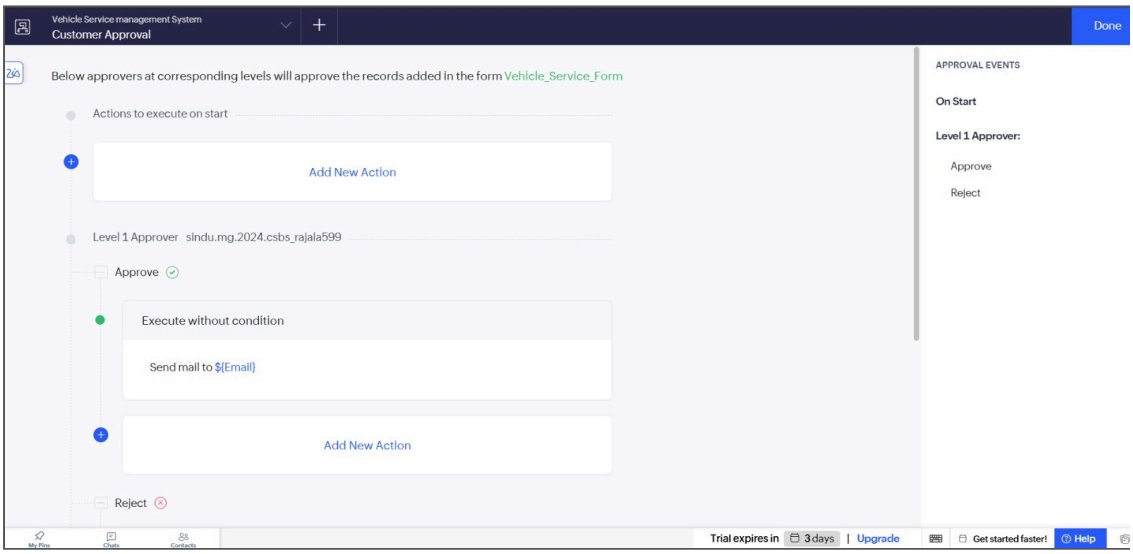


Figure 5.8: Approval Process “Customer Approval” on Vehicle_Service_Form

In live mode the Approvals section contains Pending Tasks and Completed Tasks. Figure 5.9 shows Pending Tasks with the message “There are no pending approvals available for this application” and the footer “Showing recent 0 tasks”.

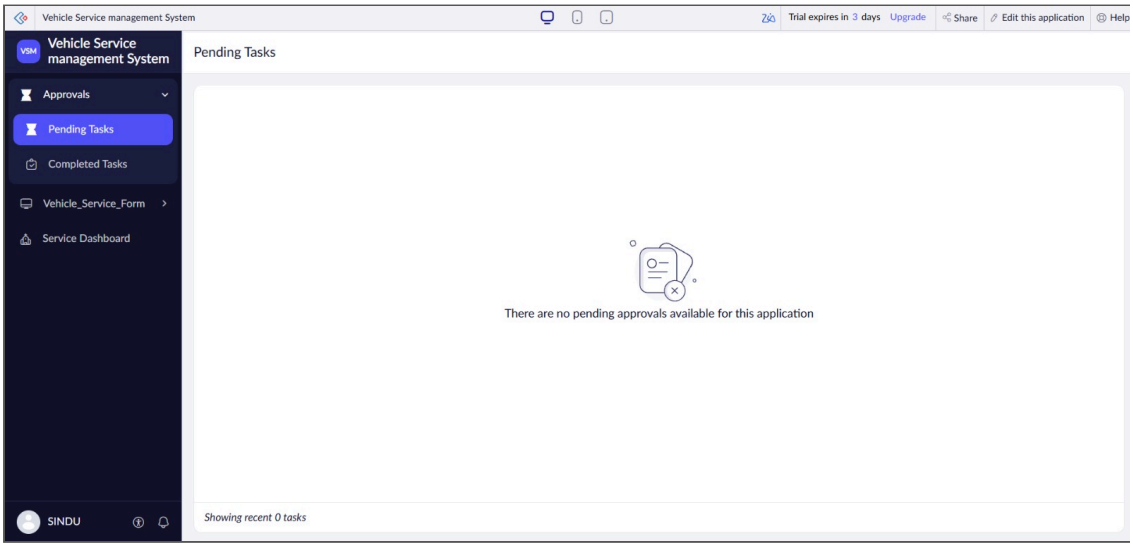


Figure 5.9: Approvals – Pending Tasks (No Pending Approvals)

Figure 5.10 shows Completed Tasks. One task is listed, dated Nov 01 2026, Sunday, at 22:12:53 hours: “Your ‘Vehicle_Service_Form’ request has been reviewed”, with Customer_Name varsha soundararajan, Contact_Number +917695987240 and her e-mail address. The footer reads “Showing recent 1 tasks”.

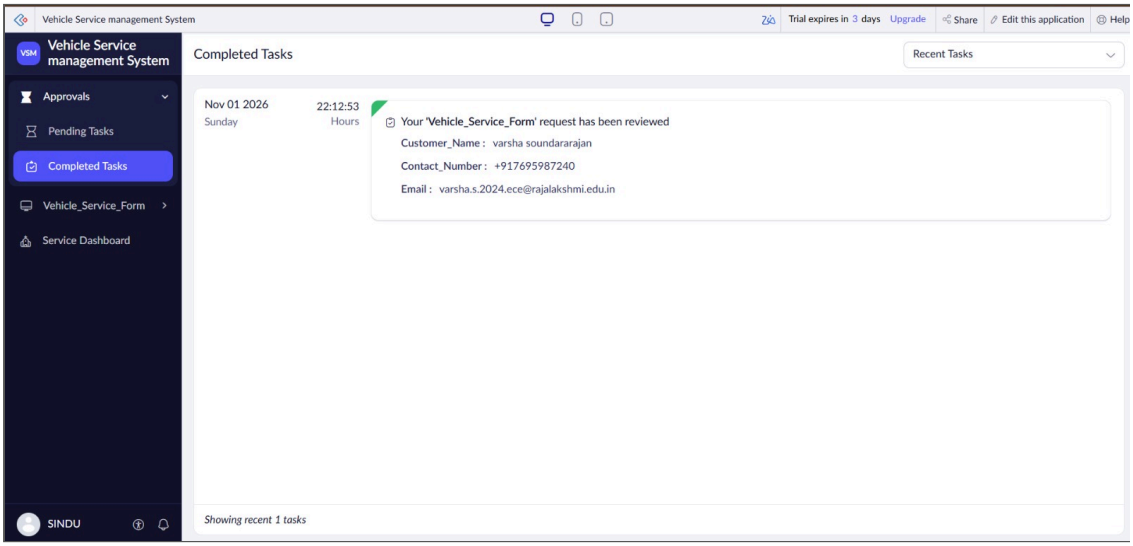


Figure 5.10: Approvals – Completed Tasks

Only one of the five bookings appears in the approval lists. varsha soundararajan is the last record in the reports, which is consistent with the approval process having been created after the first four bookings had been entered; an approval process applies to records added after it is enabled. This is recorded as observation 6 in Table 6.4.

5.6 Workflows and Automation

The Workflow section of the builder offers Form workflows, Schedules, Approvals, Payments, Blueprints, Batch workflows and Functions. The Form workflows tab (Figure 5.11) lists Vehicle_Service_Form under *Forms with workflows*; filtered by the event Created → Successful Form Submission it shows one workflow, **Notify customer**, with 1 Action.

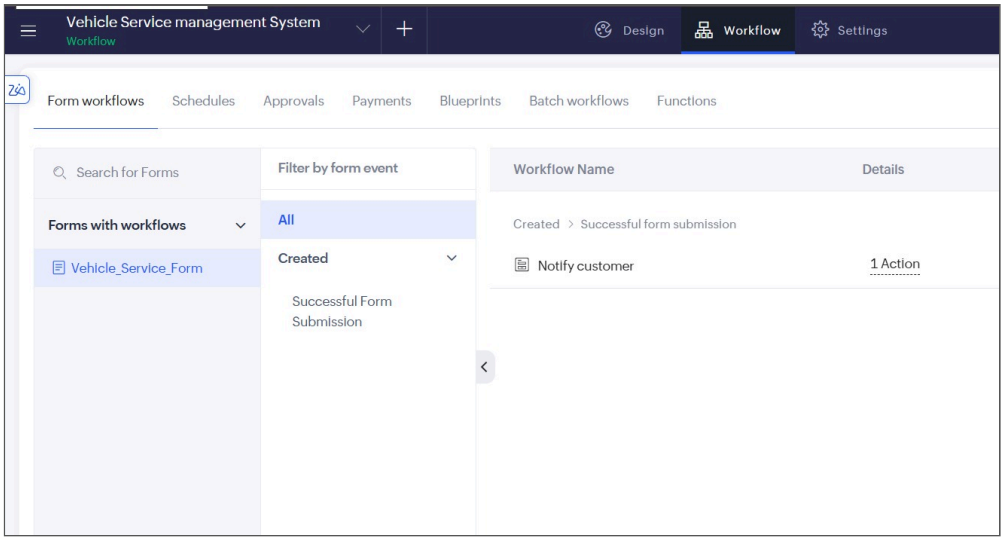


Figure 5.11: Form Workflows Page Showing the Workflow Notify customer

Table 5.2: Workflows Implemented

Workflow	Type	Trigger Shown	Actions
Notify customer	Form	Created → Successful form submission	1 – Send mail to ${Email}
Customer Approval	Approval	Records added in Vehicle_Service_Form	On Approve: Send mail to ${Email}

5.6.1 Workflow Action: Send Email of Notify customer

Opening the action shows the path *Vehicle_Service_Form › Created › Successful form submission › Notify customer* and the note “Actions will be executed on successful form submission”. The action is *Execute without condition → Send mail to ${Email}*. The Send email dialog (Figure 5.12) shows the sender **From: zoho.loginuserid**, the recipient **To: ${Email}** (the value of the Email field of the submitted record), options to add Cc, Bcc and Reply-to, the **Subject: Record Added**, and the start of the message. A notice states that, to improve deliverability, e-mails will be sent through Zoho’s domain if an unauthenticated or public domain is used.

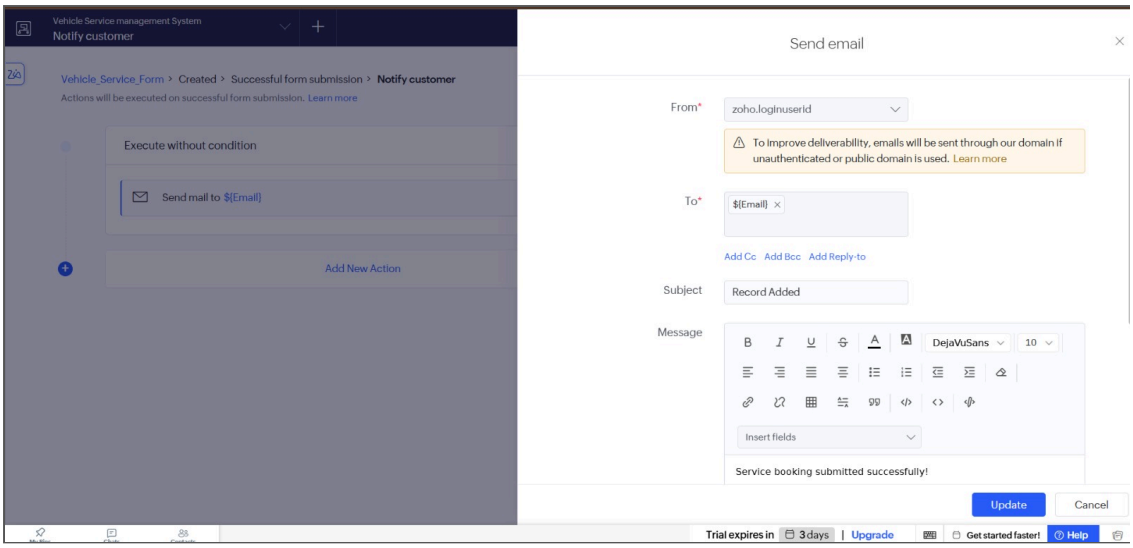


Figure 5.12: Send Email Action of Notify customer – Sender, Recipient and Subject

Scrolling down the same dialog (Figure 5.13) shows the rich-text editor with an *Insert fields* drop-down and the message body “Service booking submitted successfully!”. The option **Include user submitted data** is not ticked, and the **Content type** is Plain Text. The customer therefore receives a short confirmation without the booking details; observation 4 in Table 6.4 suggests inserting the vehicle number and service status into the message.

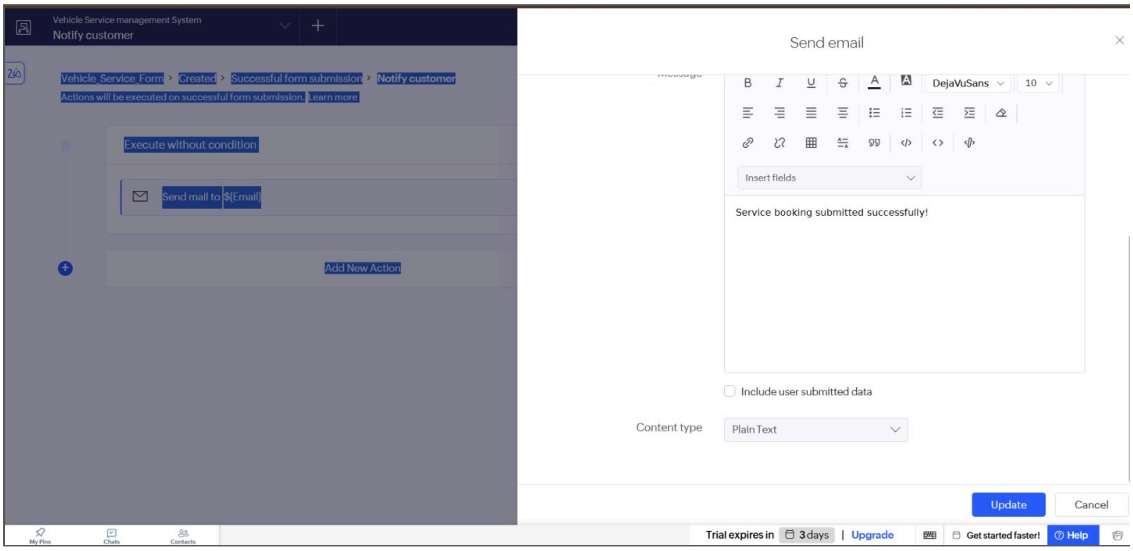


Figure 5.13: Send Email Action of Notify customer – Message and Content Type

5.7 Deluge Scripting and Functions

The present version of the application does not contain Deluge scripts or functions: the e-mail and approval actions are configured visually, which is one of the strengths of the low-code approach. Two scripts are nevertheless proposed in Appendix C, because the validation review in Chapter 6 shows that the free-text fields need checks that the field

properties cannot express. They are (1) an *On Validate* script that normalises and checks the vehicle number against the registration-mark format and makes the e-mail address required, and (2) a function that counts bookings by service status, which can feed the dashboard panels so that they always agree with the reports. These scripts are documented as proposals and are not claimed as part of the tested application.

5.8 Notifications

Table 5.3: E-mail Routes of the Application

Route	Sender	Recipient	Subject / Message	Evidence
Booking confirmation (Notify customer)	zoho.loginuserid	${Email} of the booking	“Record Added” / “Service booking submitted successfully!” (Plain Text)	Configured (Figures 5.12, 5.13)
Booking approved (Customer Approval)	Not visible	${Email} of the booking	Not visible	Configured (Figure 5.8)

No received e-mail is shown in the supplied material, so delivery of both routes is recorded as configured but not observed (Table 6.2).

5.9 Integration Configuration

No connection to another Zoho service or external system is configured in the present version. The integration opportunities identified during the work — Zoho Books for invoicing, Zoho Desk or Zoho CRM for customer history, and SMS or WhatsApp notification through a connection and *invokeurl* — are listed in Section 9.2.

5.10 User Management and Role-Based Access

The live-mode screenshots show a single signed-in user, SINDU, who is also the Level 1 approver. Users, roles, permission profiles and data-sharing rules are not shown, so role-based access is recorded as not configured. A role design for Advisor, Manager and Customer is proposed in Section 9.2.

CHAPTER 6
TESTING AND OBSERVATIONS

6.1 Testing Strategy

Testing was carried out by using the application in live mode as the owner, who also acts as advisor and approver, and checking the result in the reports, the approval lists and the dashboard. The stored records were then reviewed field by field against the validation rules expected of a service booking (Section 6.4), and the counts shown on different screens were reconciled with each other (Section 6.5). Only results supported by the supplied screenshots are recorded as observed; features that are configured but whose operation is not shown are listed separately in Table 6.2.

6.2 Test Cases and Results

Table 6.1: Test Cases and Observed Results

ID	Test	Evidence	Observed Result
T1	Open Vehicle_Service_Form in live mode	Figure 5.2	Customer Details and Vehicle Details fields displayed; Customer_Name marked mandatory; +91 code and fuel options shown
T2	Customer_Name is mandatory	Figure 5.3	Mandatory option ticked in Field Properties
T3	Bookings stored after submission	Figures 5.4, 5.5	Five bookings listed; footer “Showing 5 of 5” in both reports
T4	Brand and fuel stored from controlled lists	Figure 5.4	Brands Maruti Suzuki, Hyundai, Tata; fuel types Petrol, Diesel, Electric only
T5	Approval request reviewed	Figure 5.10	One task completed on 01 Nov 2026 at 22:12:53 for varsha soundararajan
T6	No booking left pending	Figure 5.9	“There are no pending approvals available for this application”
T7	Notify customer workflow configured	Figures 5.11–5.13	One action under Created → Successful form submission; Send mail to ${Email}, subject “Record Added”
T8	Approve action configured	Figure 5.8	Level 1 approver named; Send mail to ${Email} on Approve
T9	Pie chart agrees with records	Figure 5.7	Booked : In Progress : Completed ≈ 3 : 1 : 1 = 5 records
T10	KPI panels agree with records	Figures 5.6, 5.4	Not agreed: Total Customer 4 (2 + 1 + 1) against 5 records
T11	Embedded report on dashboard	Figure 5.7	Vehicle_Service_Form Report embedded with 5 records

ID	Test	Evidence	Observed Result
T12	Navigation menu	Figure 5.1	Approvals, Vehicle_Service_Form (form and two reports) and Service Dashboard reachable from the menu

Table 6.2: Configured Features Not Fully Verified by the Supplied Material

Feature	What Is Shown	What Is Not Shown
Notify customer e-mail	Send mail action with sender, recipient, subject and message	An e-mail received in a customer mailbox
Approval e-mail on Approve	Send mail to ${Email} under Approve	The e-mail content and its delivery
Reject branch	Reject heading in the approval process	Any action on Reject, or a rejected task
Service Details fields	Section heading in the builder and status values in the chart	The fields themselves and their properties
Dashboard KPI formulas	Four KPI values	The criteria behind each panel (e.g. which status “Service Booking” counts)
Access control	A single signed-in user	Other users, roles or permission profiles

6.3 Sample Data Used in Testing

The five bookings stored in the application are listed in Table 6.3 as shown in the Vehicle_Service_Form Report (Figure 5.4). Contact numbers and e-mail addresses are omitted here; they appear in the figures.

Table 6.3: Booking Records Observed in the Reports

No.	Customer Name	Vehicle Number	Brand	Model	Fuel Type
1	Vanitha S	TN14GH3456	Maruti Suzuki	Swift	Electric
2	Yuvasri K	TN09EF9012	Hyundai	i20	Petrol
3	Sahana R	TN22CD5678	Tata	Nexon	Diesel
4	Sindu MG	TN214567	Maruti Suzuki	SEDANS	Diesel
5	varsha soundararajan	TN29GT2930	Hyundai	venue	Petrol

6.4 Data Validation Review

Each field of the stored records was checked against the rule a service centre would expect. Table 6.4 lists the rule, whether the application enforces it, and the result on the five records. The vehicle number rule follows the registration-mark format of the

Central Motor Vehicles Rules [7]: a two-letter state code, a one- or two-digit RTO code, an optional letter series and a four-digit number (for example TN 14 GH 3456).

Table 6.4: Field-Level Validation of the Stored Records

Field	Expected Rule	Enforced?	Result on 5 Records	Verdict
Customer_Name	Required; proper case	Mandatory: Yes. Case: No	5/5 present; 1 entered in lower case (varsha soundararajan)	Partly
Contact_Number	Valid number with country code	Yes (Phone type, +91)	5/5 are +91 followed by 10 digits	Pass
Email	Valid address; required for notification	Format: Yes. Required: No	5/5 valid; all on rajalakshmi.edu.in	Partly
Vehicle_Number	Registration-mark format; unique	No (Single Line)	4/5 match; TN214567 has no RTO-code/series split and does not match; no duplicates	Fail (1)
Vehicle_Brand	From a fixed list	Yes (Drop Down)	5/5 from the list	Pass
Vehicle_Model	A model of the chosen brand	No (Single Line)	“SEDANS” is a body type, not a model; inconsistent case (venue, Swift)	Fail (1)
Fuel_Type	One option; plausible for the model	Option: Yes. Plausibility: No	5/5 valid options; Swift recorded as Electric should be confirmed with the customer	Check (1)

The review shows that the fields whose type or list is controlled by the platform (phone, e-mail, brand, fuel) hold clean data, while every error found is in a free-text field (Vehicle_Number, Vehicle_Model) or in a relationship between two fields (model and fuel type). This supports the design point in Section 4.7: controlling the input is more effective than correcting it later. The proposed On Validate script in Appendix C.3 would have rejected TN214567 with a message, and a model drop-down that depends on the brand would prevent entries such as “SEDANS”.

6.5 Cross-Screen Consistency Check

A dashboard is only useful if its numbers agree with the records. Table 6.5 reconciles the counts shown on each screen.

Table 6.5: Reconciliation of Counts Across Screens

Screen	Figure	Count Shown	Agrees with 5 Records?
Vehicle_Service_Form Report	5.4	5 records	Yes
Customer report	5.5	5 records	Yes

Screen	Figure	Count Shown	Agrees with 5 Records?
Dashboard embedded report	5.7	5 records	Yes
Dashboard pie chart	5.7	Booked ≈ 3, In Progress ≈ 1, Completed ≈ 1 (total 5)	Yes
Dashboard KPI panels	5.6	Total Customer 4; Service Booking 2; In progress 1; Completed 1 (2 + 1 + 1 = 4)	No – one record short
Approvals	5.9, 5.10	0 pending, 1 completed	Covers 1 of 5 (see Section 5.5)

The KPI panels are internally consistent (2 + 1 + 1 = 4) and the pie chart shows three Booked against the panels' two, so the most likely explanation is that the KPI screenshot was captured before the fifth booking (varsha soundararajan, Petrol, status Booked) was added, or that the panels use a cached or filtered count. Either way, the panels should be re-checked; computing them with the function in Appendix C.4 would keep them equal to the report.

6.6 Observations and Improvements

Table 6.6: Observations from Testing and Suggested Improvements

No.	Observation	Suggested Improvement
1	Vehicle_Number is free text; TN214567 does not follow the registration-mark format.	Add the On Validate script of Appendix C.3 and enable "No duplicate values".
2	Vehicle_Model holds "SEDANS", a body type, and mixes cases (venue, Swift).	Replace with a drop-down filtered by Vehicle_Brand, or a Model lookup form.
3	Fuel type is not checked against the model (Swift recorded as Electric).	Store the fuel types offered for each model and warn on an unusual combination.
4	The Notify customer e-mail has the generic subject "Record Added" and no booking details; "Include user submitted data" is not ticked.	Use a subject such as "Service Booking Confirmed – ${Vehicle_Number}" and insert the vehicle and status fields.
5	Email is not mandatory although both e-mails depend on it.	Make Email mandatory or skip the mail when it is empty.
6	Only one of the five bookings went through the approval process.	Enable the approval before data entry, or re-submit the earlier records for approval.
7	No action is visible on Reject.	Send a rejection e-mail with the reason and set the status accordingly.
8	Dashboard KPI panels show 4 customers against 5 records.	Re-check the panel criteria; compute them from the form with a function (Appendix C.4).

No.	Observation	Suggested Improvement
9	The Customer report repeats the Vehicle_Service_Form Report.	Show only customer columns or group bookings by customer.
10	Customer_Name is not normalised to proper case.	Use the field's case setting or convert the name in the validation script.
11	The account has only one user, who is also the approver.	Invite users for Advisor and Manager roles and repeat the approval tests.

6.7 Requirements Traceability

Table 6.7 relates each functional requirement of Section 3.2 to the tests and figures that support it and states how far it was observed.

Table 6.7: Traceability of Functional Requirements to Evidence

ID	Requirement (short)	Evidence	Status
FR1	Customer details	T1, T2, T3; Figures 5.2–5.4	Observed
FR2	Vehicle details	T1, T4; Figures 5.2, 5.4	Observed
FR3	Service details and status	T9; Figures 5.3, 5.7	Status values observed; fields not shown
FR4	Confirmation e-mail	T7; Figures 5.11–5.13	Configured; delivery not shown
FR5	Approval with Pending and Completed Tasks	T5, T6, T8; Figures 5.8–5.10	Observed for one booking
FR6	Reports with search and add	T3; Figures 5.4, 5.5	Observed
FR7	Dashboard KPIs and chart	T9, T10, T11; Figures 5.6, 5.7	Observed; KPI total differs
FR8	Reject wrongly formatted entries	Section 6.4	Partly: typed fields only

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Status

Zoho Creator provides a Development, Stage and Production environment model, in which an application is changed in Development, tested in Stage and published to Production without affecting the live application [2]. The supplied screenshots show the application being used in live mode from the builder of a trial account ("Trial expires in 3 days" on 01 November 2026), with the Share and Edit this application options in the header. The Environments page and a published version are not shown, so deployment to Stage and Production is recorded as not evidenced. The application is opened for the demonstration from the owner's account, and access is shared with the examiner at that time.

7.2 Security Features

Table 7.1: Security Controls Observed

Control	Evidence
Platform sign-in	Application opened by the signed-in user SINDU (every live-mode figure)
Named approver	Only sindu.mg.2024.csbs_rajala599 is listed as Level 1 approver (Figure 5.8)
Typed input	Name, Phone (+91) and Email field types; brand drop-down; fuel radio (Figure 5.2)
Mandatory field	Customer_Name mandatory (Figure 5.3)
Sender policy	Mail sent from zoho.loginuserid; platform notice on sending through Zoho's domain for deliverability (Figure 5.12)
Review trail	Completed task with date, day and time of review (Figure 5.10)

Customer phone numbers and e-mail addresses are personal data. Before real use, access to the reports should be limited by role, and the reports shared with the manager should hide the contact columns that the manager does not need.

7.3 Governance and Administration

The owner manages the form, reports, workflow and approval from the builder (Design, Workflow and Settings tabs, Figure 5.11). The Completed Tasks list keeps a dated record of each approval decision, which serves as a basic audit trail for bookings. Users, roles and connection access would be managed from the Settings tab when more users are added.

7.4 Risks and Mitigation

The risks in Table 7.2 arise from the way the application was built and tested. Each has a mitigation that can be carried out in the platform without changing the design.

Table 7.2: Risks and Mitigation

Risk	Effect	Mitigation
The account is a trial that expires (3 days shown on 01 November 2026)	The application may become unavailable shortly after the demonstration	Upgrade the plan, or export the records and reports before the trial ends
Free-text vehicle number and model	Wrong or duplicate vehicles in the records	Validation script, no-duplicate rule and model drop-down
Email not mandatory	Customer not notified	Make Email mandatory
Dashboard KPI differs from the records	Wrong decisions by the manager	Re-check panel criteria; compute from the form
Only one user, who is also the approver	No separation of duties; access rules untested	Create Advisor and Manager users and test each role
E-mail delivery not shown	Customers may not receive confirmations	Submit a test booking to a real mailbox and keep the e-mail as evidence

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The application provides a complete path from booking to status reporting: the advisor enters a booking in Vehicle_Service_Form; the booking is stored and listed in two reports; a confirmation e-mail is configured to go to the customer; the approver reviews the booking under Approvals and the customer is e-mailed on approval; and the Service Dashboard summarises the bookings by status with KPI panels, a pie chart and an embedded report.

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective	Outcome	Status
Form for customer, vehicle and service details	Vehicle_Service_Form with three sections and typed fields (Figures 5.2, 5.3)	Achieved
Approval process for bookings	Customer Approval with a Level 1 approver; one completed task, none pending	Achieved (one booking)
E-mail notification to the customer	Send mail on submission and on approval configured	Configured; delivery not shown
Reports and dashboard	Two list reports; dashboard with four KPI panels, pie chart and embedded report	Achieved; KPI total to be corrected
Documentation and validation	Field-level validation review, cross-screen reconciliation and 11 improvements	Achieved

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Aspect	Existing (Manual) System	Proposed Application
Data storage	Job cards and spreadsheets	One form with sections in one application
Booking review	No tracking	Approval process with Pending and Completed Tasks
Customer notification	Phone calls when remembered	Send mail on submission and on approval
Summaries	Prepared by hand	Dashboard and reports from stored records

Aspect	Existing (Manual) System	Proposed Application
Data validation	None	Typed fields, list values and a mandatory name; further rules proposed

8.4 Advantages and Limitations

Advantages.

- The whole application was built visually: form, reports, dashboard, approval and e-mail action, with no code.
- A booking is entered once and reused by the reports, the dashboard and the approval list.
- The approval list gives a dated trail of who reviewed which booking and when.
- Hosting, sign-in and mail delivery are provided by the platform.

Limitations.

- The application runs on a trial account, so it depends on the trial or an upgrade to remain available.
- Free-text fields allow invalid vehicle numbers and models (Section 6.4).
- The dashboard KPI total differs from the records (Section 6.5).
- E-mail delivery and the Reject branch are not shown.
- Only one user exists, so role-based access has not been configured or tested.
- Low-code platforms limit customisation of pages and logic, and complex rules still need scripting [5], [6].

8.5 Discussion

The results show that the main business path works: a booking is stored, listed, reviewed and reported on, and the pie chart and reports agree on five records. The strongest evidence is for the parts that every user sees — the form, the reports, the approval lists and the dashboard — and the weakest is for actions whose effect is outside the application, namely the e-mails.

The validation review was the most useful part of the documentation. It showed that the platform’s field types already keep four of the seven fields clean, and that every error found sits in a field left as free text. The lesson is general to low-code work: the quickest field to add (Single Line) is also the one that needs the most care afterwards. The corrections in Table 6.6 are small, each can be made in the builder or with a short script, and each can be re-tested with a single new booking.

8.6 Supporting Evidence of the Results

The results above rest on thirteen screenshots placed in Chapter 5: one of the menu (Figure 5.1), two of the form (Figures 5.2, 5.3), two of the reports (Figures 5.4, 5.5), two of the dashboard (Figures 5.6, 5.7), three of the approval process and lists (Figures 5.8–5.10) and three of the workflow and its e-mail action (Figures 5.11–5.13). The reconciliation in Table 6.5 uses five of them together and shows that the reports, the embedded report and the pie chart agree, while the KPI panels need correction.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

This project built a vehicle service management application on Zoho Creator. One form with three sections captures customer, vehicle and service details; two reports and a dashboard give a live view of the bookings; an approval process lets a named approver review each booking; and a form workflow and an approval action send e-mails to the customer.

The supplied screenshots show bookings being stored and listed, an approval being completed with a dated record, the e-mail actions configured, and a dashboard whose chart agrees with the records. The report has also documented and validated the data: it found one invalid vehicle number, one body type stored as a model, one fuel entry to be confirmed and a KPI total one short of the records, and it proposed a correction for each. The work shows that a manual service-booking process can be moved to a connected application quickly using low-code tools, and that validation rules deserve as much attention as the screens themselves.

9.2 Future Enhancements

- Add the On Validate script for the vehicle number, make Email mandatory and enable no-duplicate values on Vehicle_Number.
- Replace Vehicle_Model with a brand-dependent drop-down and check fuel type against the model.
- Separate Customer and Vehicle forms linked to bookings by lookup fields, so a returning customer is stored once.
- Add a rejection e-mail, include the booking details in both e-mails, and add an e-mail when the status becomes Completed.
- Compute the dashboard panels from the form with a function and add a chart of bookings by brand and by date.
- Create Advisor, Manager and Customer roles with permission profiles, and a customer portal to track the service status.
- Add a scheduled workflow that reminds customers of the next periodic service.
- Integrate with Zoho Books for invoices and with an SMS or WhatsApp service through a connection.

Table 9.1 ranks the enhancements by priority and links each to the observation that led to it.

Table 9.1: Priority of Future Enhancements

Enhancement	Benefit	Linked Observation	Priority
Vehicle number validation and no-duplicate rule	Clean, unique vehicle records	Observation 1	High
Correct dashboard KPI panels	Manager sees true counts	Observation 8	High
Mandatory Email and richer e-mail content	Customers reliably informed with details	Observations 4, 5	High
Brand-dependent model list; fuel check	Consistent vehicle data	Observations 2, 3	Medium
Reject action and approval for all bookings	Complete review trail	Observations 6, 7	Medium
Roles, users and customer portal	Separation of duties; self-service status	Observation 11	Medium
Distinct Customer report; name case	Clearer reports	Observations 9, 10	Low

9.3 Learning Outcomes and Career Relevance

Developing this application gave practical experience in form design with sections and typed fields, report and dashboard design, approval processes, workflow-based notification and, through the validation review, in checking the quality of application data against business rules. These skills relate to roles such as citizen developer, low-code developer, business analyst and quality analyst.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Y = implemented, Partial = implemented only in part (see note), N = not implemented. The checklist reflects the application as built and tested and should be re-checked against the live application before the demonstration.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Menu: Approvals, Vehicle_Service_Form (form and two reports), Service Dashboard	4.5, 5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Partial	Name, Phone, Email, Single Line, Drop Down, Radio visible (6); Service Details fields not visible	4.3, 5.2	Fig. 5.2, 5.3
II	Lookup field (relationship between forms)	N	Single form; lookup proposed	4.2, 9.2	—
II	Subform (parent-child data)	N	Not used	4.2	—
II	Field validations / layout rules	Partial	Mandatory Customer_Name; typed phone and e-mail; no pattern or duplicate rule	4.7, 6.4	Fig. 5.3; Table 6.4
II	List report with filter, grouping, custom action	Partial	Two list reports with search and add; no filter, grouping or custom action	5.3	Fig. 5.4, 5.5
II	Kanban / Calendar / Timeline / Map report	N	Not used	—	—
II	Pivot table / chart report	Partial	Pie chart of service status on the dashboard; no separate chart report	5.4	Fig. 5.7
II	Page / dashboard built with UI Builder	Y	Service Dashboard: four KPI panels, pie chart, embedded report	5.4	Fig. 5.6, 5.7
II	Deluge – form events (validate / success / field actions)	Partial	Successful-submission event used by a visual Send mail action; no Deluge; On Validate script proposed	5.6, 5.7	Fig. 5.11; C.3
II	Deluge – functions, fetch, loops, collections	N	Not in the application; function proposed in C.4	5.7	C.4
II	Email / in-app notifications	Y	Send mail on submission (Notify customer) and on approval	5.6.1, 5.8	Fig. 5.8, 5.12, 5.13
II	Form workflow	Y	Notify customer on Vehicle_Service_Form	5.6	Fig. 5.11

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Scheduled workflow	N	Not used; service reminder proposed	9.2	—
II	Approval process	Y	Customer Approval; Pending and Completed Tasks	5.5	Fig. 5.8–5.10
II	Blueprint	N	Not used	—	—
II	Batch / report workflow	N	Not used	—	—
II	Testing & debugging evidence	Y	Test cases T1–T12, validation review, reconciliation, observations	6	Tables 6.1–6.7
III	Integration with a Zoho service	N	Not configured	5.9	—
III	Connection / REST API call (invokeurl)	N	Not configured	5.9	—
III	Webhook or Creator REST API	N	Not used	—	—
III	Users, roles and permission profiles	N	Single user; roles proposed	5.10	—
III	Record-level access criteria / customer portal	N	Not configured; portal proposed	9.2	—
III	Zia / AI feature	N	Not used	—	—
III	Analytics (charts / Zoho Analytics)	Partial	KPI panels and pie chart on the dashboard	5.4	Fig. 5.6, 5.7
IV	Performance optimisation applied & measured	N	Nothing measured	—	—
IV	Deployment across environments	N	Live mode of the trial shown; Stage/Production not shown	7.1	—
IV	Security controls (encryption, OAuth, permissions)	Partial	Sign-in, named approver, typed input	7.2	Table 7.1
IV	Governance (audit trail, administration)	Partial	Dated Completed Tasks as review trail; no audit log shown	7.3	Fig. 5.10
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not used	—	—
V	Low-code limitations & spectrum analysis	Partial	Limitations in Section 8.4; validation lessons in 8.5	8.4, 8.5	—
—	Mobile app / portal access	N	Device preview icons in the header; not tested on mobile	4.5	—

Total features implemented: 6 fully, 9 partly, 17 not implemented, out of 32.

APPENDIX B – COURSE OUTCOME MAPPING

The self-assessed level (1 – Low, 2 – Medium, 3 – High) is left for the student; the examiner may revise it.

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2		
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5, 6; App. C		
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 5.9, 5.10, 7.2		
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7		
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 8.4, 8.5, 9.3		

Note: CO3 is evidenced mainly by design and proposals, because no integration or custom role is configured in the present version (Table A.1). CO5 also refers to AI-assisted development; no Zia feature is used, so its evidence is limited to the discussion in Sections 8.4, 8.5 and 9.3.

APPENDIX C – WORKFLOW CONFIGURATION AND PROPOSED DELUGE CODE

C.1 and C.2 record the configuration of the actions as shown in the builder (Figures 5.8, 5.12 and 5.13); they are visual actions, not scripts. C.3 and C.4 are Deluge scripts **proposed** in this report to close the validation and consistency gaps found in Chapter 6. They are not part of the tested application, and the field link name of the status field in C.4 is assumed and must be matched to the actual field.

C.1 Workflow "Notify customer" – Send mail action (configuration)

```
Workflow  : Notify customer
Path      : Vehicle_Service_Form > Created > Successful form submission
            > Notify customer
Condition : Execute without condition
Action    : Send mail
From      : zoho.loginuserid
To        : ${Email}
Subject   : Record Added
Message   : Service booking submitted successfully!
Content   : Plain Text   (option "Include user submitted data" not ticked)
```

C.2 Approval "Customer Approval" (configuration)

```
Approval            : Customer Approval
Form                : Vehicle_Service_Form (records added)
Actions on start    : none
Level 1 Approver    : sindu.mg.2024.csbs_rajala599
  On Approve        : Execute without condition -> Send mail to ${Email}
  On Reject         : (no action visible)
```

C.3 Proposed On Validate script for Vehicle_Service_Form

```
// Script ID : DS-P1 (proposed)
// Location  : Vehicle_Service_Form -> Created or Edited
//             -> Validations on form submission
// Purpose   : Reject invalid vehicle numbers, duplicate vehicles
//             and bookings without an e-mail address

// 1. Normalise the vehicle number: upper case, no spaces or hyphens
vehNo = ifnull(input.Vehicle_Number, "").toUpperCase();
vehNo = vehNo.replaceAll(" ", "").replaceAll("-", "");

// 2. Registration mark: state code, RTO code, optional series, 4 digits
if(!vehNo.matches("^[A-Z]{2}[0-9]{1,2}[A-Z]{0,3}[0-9]{4}$"))
{
    alert "Enter a valid vehicle number, for example TN14GH3456";
    cancel submit;
}
input.Vehicle_Number = vehNo;

// 3. The customer e-mail is needed for both notifications
if(input.Email == null || input.Email == "")
{
    alert "E-mail is required to send the booking confirmation";
    cancel submit;
}

// 4. Prevent the same vehicle being booked twice (other records only)
if(input.ID == null)
{
    dup = Vehicle_Service_Form[Vehicle_Number == vehNo];
}
else
{
    dup = Vehicle_Service_Form[Vehicle_Number == vehNo && ID != input.ID];
}
if(dup.count() > 0)
{
    alert "A booking already exists for vehicle " + vehNo;
    cancel submit;
}
```

With this script, record 4 of Table 6.3 (TN214567) would be rejected with the alert message, while the other four numbers would pass and be stored without spaces in upper case.

C.4 Proposed function countBookingsByStatus()

```
// Function ID : DF-P1 (proposed)
// Location    : Workflow -> Functions -> Default
// Purpose     : Return the number of bookings with a given status so that
//               the dashboard KPI panels always match the stored records
// Note        : Service_Status is an assumed field link name

int countBookingsByStatus(string status)
{
    if(status == "All")
    {
        return Vehicle_Service_Form[ID != 0].count();
    }
    return Vehicle_Service_Form[Service_Status == status].count();
}

// Usage on the Service Dashboard page:
//   Total Customer   -> thisapp.countBookingsByStatus("All")
//   Service Booking  -> thisapp.countBookingsByStatus("Booked")
//   In progress      -> thisapp.countBookingsByStatus("In Progress")
//   Completed status -> thisapp.countBookingsByStatus("Completed")
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Sign in to Zoho Creator and open **Vehicle Service management System** from the Applications area. The menu on the left contains Approvals, Vehicle_Service_Form and Service Dashboard (Figure 5.1).

D.2 Instructions for the Service Advisor / Administrator

1. Open **Vehicle_Service_Form → Vehicle_Service_Form**.
2. Under Customer Details, enter the customer's first and last name (mandatory), the contact number (the +91 code is pre-selected) and the e-mail address.
3. Under Vehicle Details, enter the vehicle number in the format TN14GH3456, choose the brand, type the model and select the fuel type.
4. Under Service Details, enter the service required and set the status to Booked.
5. Click **Submit**. The booking is stored, a confirmation e-mail "Record Added" is sent to the customer and an approval request is created.
6. Open **Vehicle_Service_Form Report** or **Customer report** to search, view or edit bookings; update the status to In Progress and then Completed as the job proceeds.

D.3 Instructions for the Approver

1. Open **Approvals → Pending Tasks** to see the bookings waiting for review.
2. Open a task, check the customer and vehicle details, and click **Approve** or **Reject**.
3. On approval the customer is e-mailed and the task moves to **Completed Tasks**, where the date and time of review are shown.

D.4 Instructions for the Service Manager

1. Open **Service Dashboard** to see Total Customer, Service Booking, In progress and Completed status, and the pie chart of bookings by status.
2. Scroll down to the embedded report to see the individual bookings, and use the search icon to find a vehicle or customer.

The Approver and Manager instructions describe the roles as designed. The account lists only one user, who performs all roles, so these roles were not tested with separate logins (Section 5.10, Table 6.2).